Chapter 5, 6
Integrals & Techniques of Integration

MAE101 - CALCULUS

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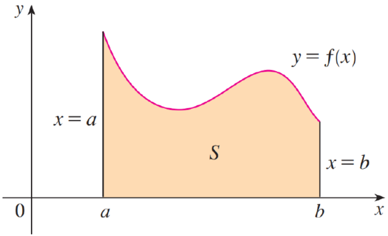
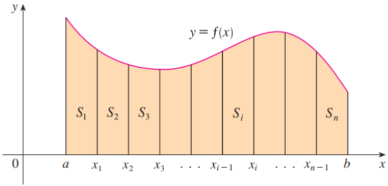
The area problem

1 Definite Integral

Find the area of the region S that lies under the curve $y = f(x)$ from $x = a$ to $x = b$.



We start by subdividing S into strips $S_1; S_2; ...; S_n$ of equal width as in Figure below



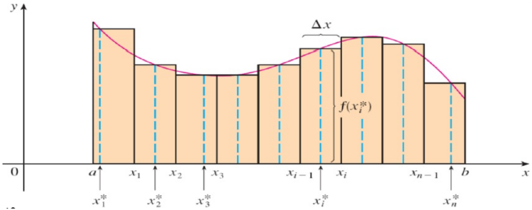
The area problem

1 Definite Integral

Let's approximate the th strip S_i by a rectangle with width Δx and height $f\left(x_i^*\right)$.
Then the area of the rectangle is $f\left(x_i^*\right)\Delta x$.
On each rectangle S_i takes any number x_i^* in the subinterval $[x_{i-1}, x_i]$.

$$S = S_1 + S_2 + \cdots + S_n$$

$$S \simeq f\left(x_1^*\right)\Delta x + f\left(x_2^*\right)\Delta x + \cdots + f\left(x_n^*\right)\Delta x = \sum_{i=1}^{n} f\left(x_i^*\right)\Delta x_i \quad \text{(Riemann Sum)}$$



The area problem

The width of each of the n strips is $\Delta x = \Delta x_i = \dfrac{b-a}{n}$

These strips divide the interval $[a, b]$ into n subintervals

$$[x_0, x_1], [x_1, x_2], [x_2, x_3], \ldots, [x_{n-1}, x_n]$$

where $x_0 = a$ and $x_n = b$.

If $x_i^* = x_i$ for any $i \in \overline{1, n}$, then $R_n := S \simeq \sum_{i=1}^{n} f(x_i)\Delta x \quad \Longrightarrow$ Right Endpoint Rule

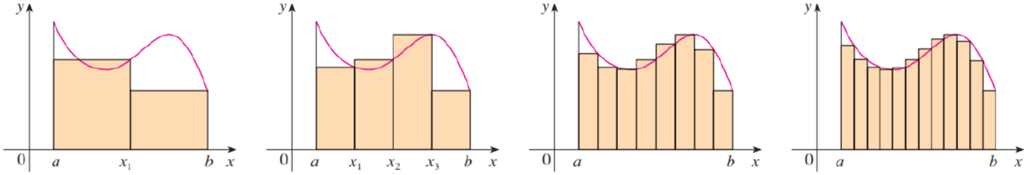
If $x_i^* = x_{i-1}$ for any $i \in \overline{1, n}$, then $L_n := S \simeq \sum_{i=1}^{n} f(x_{i-1})\Delta x \quad \Longrightarrow$ Left Endpoint Rule

If $x_i^* = \dfrac{x_{i-1} + x_i}{2}$, for any $i \in \overline{1, n}$, then $M_n := S \simeq \sum_{i=1}^{n} f\left(\dfrac{x_{i-1} + x_i}{2}\right)\Delta x \quad \Longrightarrow$ Midpoint Rule

The area problem

1 Definite Integral

Figure below shows this approximation for n = 2; 4; 8 and 12. Notice that this approximation appears to become better and better as the number of strips increases, that is, as $n \to +\infty$



(a) $n = 2$ (b) $n = 4$ (c) $n = 8$ (d) $n = 12$

Example 1. Use the Midpoint Rule with $n = 5$ to approximate $\int_1^2 \frac{1}{x} dx$.

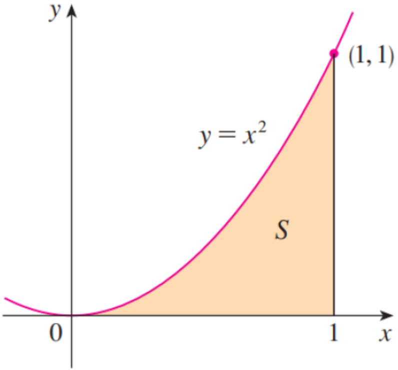
Example 2. Estimate the area under the graph of $y = x^2 - x$ from $x = 0$ to $x = 4$ using four subintervals and left endpoints.

Example

1 Definite Integral

Use rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1 . Use Right and Left Endpoint rules.

Solution. Dividing the interval $[0, 1]$ into n subintervals. So, $\Delta x = \dfrac{1}{n}$ and $x_i = \dfrac{i}{n}$ with $i = \overline{1, n}$.



$$R_n = \sum_{i=1}^{n} f(x_i) \cdot \Delta x = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}$$

$$L_n = \sum_{i=1}^{n} f(x_{i-1}) \cdot \Delta x = \frac{2n^2 - 3n + 1}{6n^2}.$$

Definite Integral

1 Definite Integral

Definition

Given a function $f(x)$ that is continuous on the interval $[a, b]$ we divide the interval into n subintervals of equal width, Δx, and from each interval choose a point, x_i^*. Then the definite integral of $f(x)$ from a to b is

$$I = \lim_{n\to+\infty}\left(\sum_{i=1}^{n} f\left(x_i^*\right)\Delta x_i\right) = \int_a^b f(x)dx.$$

Note.

1. If it does exist, we say f is integrable on $[a, b]$.
2. The definite integral $\int_a^b f(x)dx$ is a number. It does not depend on x.
3. In fact, we could use any letter in place of x without changing the value of the integral:

$$\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(r)dr$$

Properties

1 Definite Integral

We assume f and g are continuous functions on $[a, b]$.

1. $\displaystyle\int_a^b c\,dx = c(b-a)$, where c is any constant.
2. $\displaystyle\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)dx + \int_a^b g(x)dx.$
3. $\displaystyle\int_a^b cf(x)dx = c\int_a^b f(x)dx.$
4. $\displaystyle\int_a^b [f(x)-g(x)]\,dx = \int_a^b f(x)dx - \int_a^b g(x)dx.$
5. If $f(x) \geq 0$ for $a \leq x \leq b$, then $\displaystyle\int_a^b f(x)dx \geq 0.$
6. If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\displaystyle\int_a^b f(x)dx \geq \int_a^b g(x)dx.$

Properties

7. If $m \le f(x) \le M$ for $a \le x \le b$, then $m \le \dfrac{1}{b-a}\displaystyle\int_a^b f(x)dx \le M$.

   Mean Value Theorem. We call $f_{\text{ave}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)dx$ is average value of the function $f(x)$ on $[a, b]$.

8. The Fundamental Theorem of Calculus
   If $f(x)$ is continuous on $[a, b]$, and $u(x)$ is differentiable, then

$$\left(\int_a^{u(x)} f(t)dt\right)' = f(u(x)) \cdot u'(x).$$

9. If $f(x)$ is continuous on $[a, b]$, and $u(x), v(x)$ are differentiable, then

$$\left(\int_{v(x)}^{u(x)} f(t)dt\right)' = -v'(x)f(v(x)) + u'(x)f(u(x))$$

Example 1. Find the average value of $f(x) = x + 1$ over the interval $[0, 5]$.

Example 2. Find the derivative of $g(x) = \displaystyle\int_1^x \frac{1}{t^3+1}\,dt$.

Example 3. Let $F(x) = \displaystyle\int_1^{\sqrt{x}} \sin t\,dt$. Find $F'(x)$.

Example 4. Let $F(x) = \displaystyle\int_{\sqrt{x}}^{3x} t^2 \sin\left(1+t^2\right) dt$. Find $F'(x)$.

The net change theorem

The integral of a rate of change is the net change: $\int_a^b F'(x)dx = F(b) - F(a)$

Distance vs displacement

If an object moves along a straight line with position function $s(t)$, then its velocity is $v(t) = s'(t)$, the acceleration of the object is $a(t) = v'(t)$

1. Displacement $= \displaystyle\int_{t_1}^{t_2} v(t)dt = s(t_2) - s(t_1) =$ change in position.
2. Distance $= \displaystyle\int_{t_1}^{t_2} |v(t)|dt =$ total distance traveled.
3. Change in velocity $= \displaystyle\int_{t_1}^{t_2} a(t)dt = v(t_2) - v(t_1)$.

The net change theorem - Example

A particle moves along a line so that its velocity at time t is $v(t) = t^2 - t - 6$ m/s.

a. Find the displacement of the particle during the time period $1 \leq t \leq 4$.

b. Find the distance traveled during this time period.

The net change theorem - Example

A particle moves along a line so that its velocity at time t is $v(t) = t^2 - t - 6$ m/s.

a. Find the displacement of the particle during the time period $1 \le t \le 4$.

b. Find the distance traveled during this time period.

Solution.

a. The displacement is $s(4) - s(1) = \int_1^4 v(t)dt = \int_1^4 \left(t^2 - t - 6\right) dt = -\dfrac{9}{2}$

This means that the particle moved 4.5 m toward the left.

b. The distance traveled is $\int_1^4 |v(t)|dt = \int_1^3 [-v(t)]dt + \int_3^4 v(t)dt = \dfrac{61}{6} \approx 10.17m$

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2 Techniques of integration

Integration fomulars

2 Techniques of integration

1. $\int \sin ax dx = -\frac{1}{a} \cos ax + c.$
2. $\int \cos ax dx = \frac{1}{a} \sin ax + c.$
3. $\int \frac{dx}{\cos^2 x} = \tan x + c.$
4. $\int \frac{dx}{\sin^2 x} = -\cot x + c.$
5. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a} + c.$
6. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + c = -\arccos \frac{x}{a} + c.$
7. $\int a dx = ax + c.$
8. $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln\left(x + \sqrt{x^2 \pm a^2}\right) + c,\ a \neq 0.$
9. $\int \frac{1}{x} dx = \ln|x| + c.$
10. $\int x^\alpha dx = \frac{1}{\alpha + 1} x^{\alpha+1} + c.$
11. $\int e^{ax} dx = \frac{1}{a} e^{ax} + c.$
12. $\int a^x dx = \frac{a^x}{\ln a} + c.$

Suppose f is continuous on $[-a, a]$

If f is even, $[f(-x) = f(x)]$, then

$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$$

If f is odd, $[f(-x) = -f(x)]$, then

$$\int_{-a}^{a} f(x)dx = 0$$

Integration by substitution

2 Techniques of integration

Substitution Rule

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I, then

$$\int f(g(x))g'(x)dx = \int f(u)du$$

Example. Compute

1. $\displaystyle\int x^3 \cos\left(x^4 + 2\right) dx.$
2. $\displaystyle\int 2x\sqrt{x^2 + 1}dx.$

Integration by parts

2 Techniques of integration

Formula for integration by parts

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

Let $u = f(x)$ and $v = g(x)$. Then the differentials are $du = f'(x)dx$ and $dv = g'(x)dx$, so, by the Substitution Rule, the formula for integration by parts becomes

$$\int udv = uv - \int vdu$$

Example 1. Find $\int x \sin x dx$.

Example 2. Find $\int \ln x dx$.

Example 3. Find $\int e^x \sin x dx$.

Integration by parts - Solution

Example 1. We choose $f(x) = x$ and $g'(x) = \sin x$. Then $f'(x) = 1$ and $g(x) = -\cos x$. We have

$$\int x \sin x dx = f(x)g(x) - \int g(x)f'(x)dx = -x\cos x + \int \cos x dx = -x\cos x + \sin x + C.$$

Example 2. $\begin{cases} u = \ln x \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{x}dx \\ v = x \end{cases} \Rightarrow \displaystyle\int \ln x dx = x\ln x - x + C.$

Example 3. $\displaystyle\int e^x \sin x dx = \frac{e^x \sin x - e^x \cos x}{2}$ since $\cdots$

$$\begin{cases} u = e^x \\ dv = \sin x dx \end{cases} \Longrightarrow \begin{cases} du = e^x \\ v = -\cos x \end{cases} \Longrightarrow \int e^x \sin x dx = -e^x \cos x + \int e^x \cos x dx$$

$$\begin{cases} u = e^x \\ dv = \cos x dx \end{cases} \Longrightarrow \begin{cases} du = e^x \\ v = \sin x \end{cases} \Longrightarrow \int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx$$

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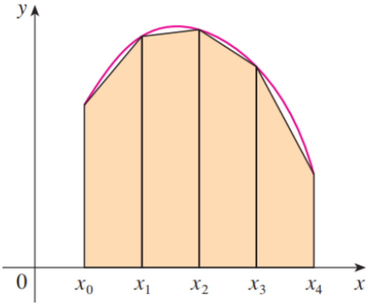
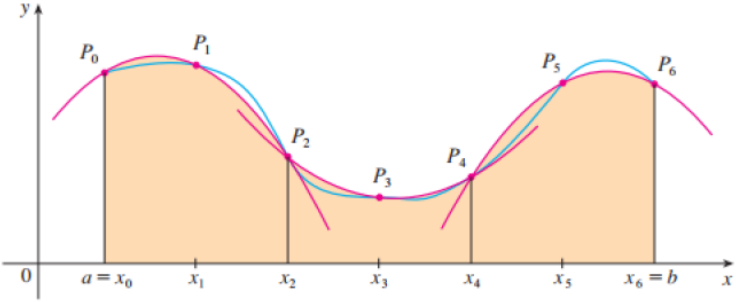
Trapezoidal Rule



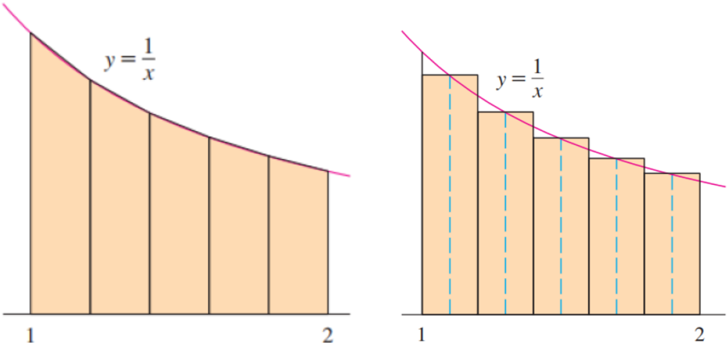
$$\int_a^b f(x)dx \approx T_n = \frac{\Delta x}{2}\left[1f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + 1f(x_n)\right]$$

Simpson's Rule



$$\int_a^b f(x)dx \approx S_n = \frac{\Delta x}{3}\left[1f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + 1f(x_n)\right]$$

Use (a) the Trapezoidal Rule with $n = 5$, (b) the Midpoint Rule with $n = 5$ and (c) Simpson's Rule with $n = 10$ to approximate the integral $\int_1^2 \frac{1}{x}dx$.



Solution

With $n = 5$, $a = 1$, and $b = 2$, we have $\Delta x = \dfrac{2-1}{5} = 0.2$, and so the

(a) Trapezoidal Rule gives

$$\int_1^2 \frac{1}{x}dx \approx T_5 = \frac{0.2}{2}[f(1) + 2f(1.2) + 2f(1.4) + 2f(1.6) + 2f(1.8) + f(2)]$$
$$= 0.1\left(\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2}\right) \approx 0.695635$$

(b) The Midpoint Rule give

$$\int_1^2 \frac{1}{x}dx \approx M_5 = \Delta x[f(1.1) + f(1.3) + f(1.5) + f(1.7) + f(1.9)]$$
$$= \frac{1}{5}\left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9}\right) \approx 0.691908$$

Solution

(c) $n = 10$, $\Delta x = 0.1$ in Simpson's Rule, we obtain

$$\begin{aligned}
\int_1^2 \frac{1}{x} dx &\approx S_{10} \\
&= \frac{\Delta x}{3}[f(1) + 4f(1.1) + 2f(1.2) + 4f(1.3) + \cdots + 2f(1.8) + 4f(1.9) + f(2)] \\
&= \frac{0.1}{3}\left(\frac{1}{1} + \frac{4}{1.1} + \frac{2}{1.2} + \frac{4}{1.3} + \frac{2}{1.4} + \frac{4}{1.5} + \frac{2}{1.6} + \frac{4}{1.7} + \frac{2}{1.8} + \frac{4}{1.9} + \frac{1}{2}\right) \\
&\approx 0.693150.
\end{aligned}$$

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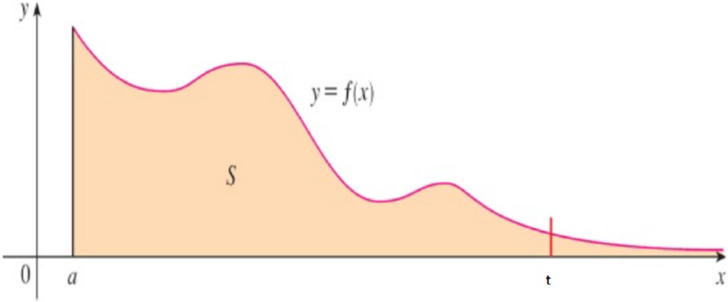
The area problem

4 Improper integral of type 1



Find the area of the region S that lies under the curve $y = f(x)$, above the -axis, and to the right of the line $x = a$.

$$S = \int_a^{+\infty} f(x)dx = \lim_{t\to+\infty} \int_a^t f(x)dx$$

Improper integral of type 1

4 Improper integral of type 1

Definition

Let $y = f(x)$ be a function defined on $[a, +\infty)$, and there exists $\displaystyle\int_a^t f(x)dx$ for every number $t \geq a$, then the integral

$$\int_a^{+\infty} f(x)dx = \lim_{t\to+\infty} \int_a^t f(x)dx$$

is called an improper integrals of type 1 .

Similarly, we have the following improper integral of type 1

$$\int_{-\infty}^{a} f(x)dx = \lim_{t\to-\infty} \int_t^a f(x)dx$$

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^{a} f(x)dx + \int_a^{+\infty} f(x)dx$$

Convergence and divergence of the improper integral of type 1

Theorem

If the improper integral $\lim_{t\to+\infty}\int_a^t f(x)dx$ is called convergent if the corresponding limit exists (as a finite number) and divergent if the limit does not exist.

Two problems with improper integral:

- Calculate the improper integrals.
- Check convergence of the improper integrals.

Example 1. Calculate and determine whether the improper integral $\int_1^{+\infty}\frac{1}{x}dx$ is convergent or divergent.

Example 1 - Solution

By definition, we have

$$\int_1^{+\infty} \frac{1}{x} dx = \lim_{t \to +\infty} \int_1^t \frac{1}{x} dx = \lim_{t \to +\infty} \ln|x| \Big|_1^t$$
$$= \lim_{t \to +\infty} (\ln t - \ln 1) = \lim_{t \to +\infty} \ln t = +\infty$$

The limit does not exist as a real number, so the improper integral $\int_1^{+\infty} \frac{1}{x} dx$ is divergent.

Example 2

4 Improper integral of type 1

Calculate and determine whether the improper integral $\int_1^{+\infty} \frac{1}{x^p} dx$, $\forall p$ is convergent or divergent.

Example 2

Calculate and determine whether the improper integral $\int_1^{+\infty} \frac{1}{x^p} dx$, $\forall p$ is convergent or divergent.

Solution.

If $p = 1$, then $\int_1^{+\infty} \frac{1}{x} dx$ is divergent (Example 1).

If $p \neq 1$, then $\int_1^{+\infty} \frac{1}{x^p} dx = \lim\limits_{t \to +\infty} \left(\frac{x^{-p+1}}{-p+1} \right) \Bigg|_{x=1}^{x=t} = \lim\limits_{t \to +\infty} \frac{1}{1-p} \left(\frac{1}{t^{p-1}} - 1 \right)$

Case 1: $p > 1$, we have $\frac{1}{t^{p-1}} \to 0$ as $t \to +\infty$, hence $\int_1^{+\infty} \frac{1}{x^p} dx = \frac{1}{p-1}$.
So, the improper integral is convergent.

Case 2: $p < 1$, we have $\frac{1}{t^{p-1}} = t^{1-p} \to +\infty$ khi $t \to +\infty$.
So, the improper integral is divergent.

Example 3

Calculate and determine whether the improper integral $\int_{-\infty}^{0} xe^x dx$ is convergent or divergent.

Example 3

4 Improper integral of type 1

Calculate and determine whether the improper integral $\int_{-\infty}^{0} xe^x dx$ is convergent or divergent.

Solution. We have

$$\int_{-\infty}^{0} xe^x dx = \lim_{t \to -\infty} \int_{t}^{0} xe^x dx = \lim_{t \to -\infty} \left(xe^x|_t^0 - \int_t^0 e^x dx \right) = -te^t - 1 + e^t = -1$$

So, the improper integral is convergent.

Notes.

1. $\lim\limits_{t \to -\infty} e^t = 0.$
2. $\lim\limits_{t \to -\infty} te^t = \lim\limits_{t \to -\infty} \dfrac{t}{e^{-t}} = \lim\limits_{t \to -\infty} \dfrac{t'}{(e^{-t})'} = \lim\limits_{t \to -\infty} \dfrac{1}{-e^{-t}} = 0.$ (L'Hôpital rule)

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5 Improper integral of type 2

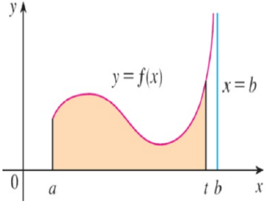
Improper integral of type 2

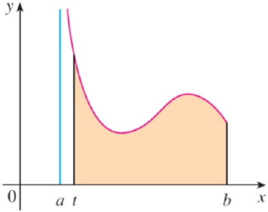
Definition

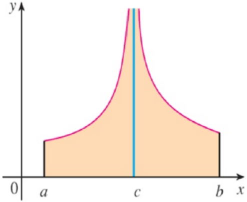
If f is continuous on $[a; b)$ and is discontinuous at b, then $\int_a^b f(x)dx = \lim_{t \to b^-} \int_a^t f(x)dx$ if this limit exists.

If f is continuous on $(a; b]$ and is discontinuous at a, then $\int_a^b f(x)dx = \lim_{t \to a^+} \int_t^b f(x)dx$ if this limit exists.

If f has a discontinuity at c, where $a < c < b$, then $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$







Example 4

5 Improper integral of type 2

Calculate and determine whether the improper integral $\int_2^5 \frac{1}{\sqrt{x-2}}dx$ is convergent or divergent.

Example 4

Calculate and determine whether the improper integral $\int_2^5 \frac{1}{\sqrt{x-2}}dx$ is convergent or divergent.

Solution. The function $f(x) = \frac{1}{\sqrt{x-2}}$ is discontinuous at $x = 2$. So

$$\begin{aligned}\int_2^5 \frac{1}{\sqrt{x-2}}dx &= \lim_{t\to 2^+}\int_t^5 \frac{dx}{\sqrt{x-2}} \\ &= \lim_{t\to 2^+} 2\sqrt{x-2}\Big|_t^5 \\ &= \lim_{t\to 2^+} 2(\sqrt{3}-\sqrt{t-2}) \\ &= 2\sqrt{3}\end{aligned}$$

Hence, the improper integral is convergent.

Example 5

5 Improper integral of type 2

Calculate and determine whether the improper integral $I = \int_0^3 \frac{dx}{x-1}$ is convergent or divergent.

Example 5

Calculate and determine whether the improper integral $I = \int_0^3 \frac{dx}{x-1}$ is convergent or divergent.

Solution. The function $f(x) = \frac{dx}{x-1}$ is discontinuous at $x = 1$. Thus,

$$I = \int_0^1 \frac{dx}{x-1} + \int_1^3 \frac{dx}{x-1} = I_1 + I_2$$

Moreover,

$$I_1 = \lim_{t \to 1^-} \int_0^t \frac{dx}{x-1} = \lim_{t \to 1^-} \ln|x-1| \Big|_0^t = \lim_{t \to 1^-} \ln|t-1| = +\infty$$

So, the improper integral I is divergent.

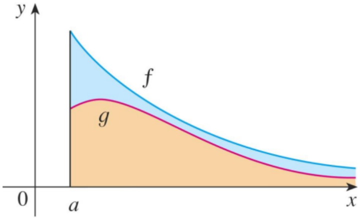
Comparison Theorem

5 Improper integral of type 2

Theorem

Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

1. If $\displaystyle\int_a^{+\infty} f(x)dx$ is convergent, then $\displaystyle\int_a^{+\infty} g(x)dx$ is convergent.
2. If is divergent $\displaystyle\int_a^{+\infty} g(x)dx$, then $\displaystyle\int_a^{+\infty} f(x)dx$ is divergent.



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Problems

6 Problems

1. Estimate the area under the graph of $y = f(x)$ using 6 rectangles and left endpoints:

   a. $f(x) = \dfrac{1}{x} + x$, $x \in [1, 4]$

   b. A table of values for f is given

x	1	2	3	4	5	6	7
$f(x)$	5	6	3	2	7	1	2

2. Repeat Prob 1 using right endpoints.

3. Use (a) the Trapezoidal Rule, (b) the Midpoint Rule, and (c) Simpson's Rule to approximate the given integral with the specified value of n.

   a. $\displaystyle\int_0^3 \sqrt{x}dx$, $n = 4$

   b. $\displaystyle\int_1^3 \frac{\sin x}{x}dx$, $n = 6$

4. Find the derivative of the function $g(x) = \displaystyle\int_0^x \sqrt{t^2+1}dt$

5. Find g'.

   a. $g(x) = \displaystyle\int_1^{x^4} \frac{1}{\cos t}dt$

   b. $g(x) = \displaystyle\int_1^{\sqrt{x}} \frac{\sin u}{x}du$

6. Find the average value of the function on the given interval

   a. $f(x) = x^2$, $[-1, 1]$

   b. $f(x) = \dfrac{1}{x}$, $[1, 5]$

7. A particle moves along a line so that its velocity at time t is $v(t) = t^2 - t - 6$ (m/s)

   a. Find the displacement of the particle during the time period $1 \leq t \leq 4$.

   b. Find the distance traveled during this time period.

8. Suppose the acceleration function and intial velocity are $a(t) = t + 3\,(m/s^2)$, $v(0) = 5\,(m/s)$. Find the velocity at time t and the distance traveled when $0 < t < 5$.

9. A particle moves along a line with velocity function $v(t) = t^2 - t(\text{ m/s})$. Find the displacement and the distance traveled by the particle during the time interval $t \in [0, 2]$.

10. Evaluate the integral $I = \displaystyle\int_0^2 x^2\sqrt{x^3 + 1}dx$.

11. Suppose $f(x)$ is differentiable, $f(1) = 4$ and $\int_0^1 f(x)dx = 5$. Find $\int_0^1 xf'(x)dx$

12. Suppose $f(x)$ is differentiable, $f(1) = 3, f(3) = 1$ and $\int_1^3 xf'(x)dx = 13$. What is the average value of f on the interval $[1, 3]$?

13. Let $f(x) = \begin{cases} -x-1 & \text{if } -3 \leq x \leq 0 \\ -\sqrt{1-x^2} & \text{if } 0 < x \leq 1 \end{cases}$. Evaluate $\int_{-3}^1 f(x)dx$

14. Calculate the improper integral.

    a. $\int_2^{+\infty} \frac{2}{x+1}dx$ b. $\int_1^{+\infty} \frac{1}{x^4}dx.$ c. $I = \int_1^{+\infty} \frac{1}{x^3}dx$

15. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

    a. $\int_1^{+\infty} \frac{dx}{(3x+1)^2}$ b. $\int_{-\infty}^0 \frac{dx}{2x-5}$ c. $\int_0^4 \frac{dx}{\sqrt{4-x}}$

Q&A

Thank you for listening!